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ORIGINAL ARTICLE

Experimental evaluation of water cooling effects on photovoltaic module performance in a hot climate

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Abstract

This study investigates the performance of a water-based cooling system for photovoltaic (PV) modules under the extreme climatic conditions of the Saharan region. The system applies intermittent water flow directly over the PV panel surface, activated at set temperatures of 35°C and 45°C, and operates at two flow rates (0.037 and 0.17 L/s). Our results demonstrate significant improvements in both temperature reduction and electrical efficiency, with temperature drops ranging from 28.7 to 33.0°C, depending on the cooling configuration. The maximal electrical efficiency of the PV panels with the water cooling system was observed at a flow rate of 0.17 m/s and a set temperature of 35°C. It increased from 17.58% (without cooling) to 21.92% (with cooling), representing a maximum improvement rate of 24.68% and an average improvement of 20.55%. Additionally, the system contributes to resource efficiency by enabling water reuse for irrigation, particularly after cooling to safe temperature levels. The intermittent water flow not only minimizes water consumption but also ensures homogeneous cooling without impacting the panel's optical properties, such as transmittance and reflectivity. Compared to existing cooling methods, the proposed system offers a cost-effective, sustainable solution that is easy to install and operate without expert intervention. This study provides valuable insights for optimizing PV performance in harsh environments, addressing both energy efficiency and water resource management.

Keywords Photovoltaic cooling · Water-based cooling · Saharan climate · Electrical efficiency · Temperature reduction · Water reuse

Introduction

The global push to diversify energy sources and reduce reliance on fossil fuels has intensified in recent years, driven largely by the urgent need to mitigate climate change (Irfan et al., 2023; Krueger et al., 2024). In this context, Algeria—an energy-dependent nation—has recognized the critical necessity of transitioning towards sustainable energy alternatives (Tambari et al., 2024). To achieve this, the Algerian government has launched an ambitious renewable energy program aimed at diversifying the country's energy mix, reducing its carbon footprint, and fostering sustainable development. The initiative focuses on accelerating the adoption of renewable energy technologies, such as solar photovoltaic (PV) systems, wind power, and other clean energy sources, which contribute to both national energy security and global climate change mitigation (Bouraiou et al., 2020, Ministry of Energy and Mining, 2023).

However, PV systems installed in the harsh Saharan climate face significant challenges. The extreme temperatures in these regions can cause fluctuations in power output and accelerate the degradation of solar modules, undermining their long-term performance and efficiency (Bouraiou et al., 2015; Chaib et al., 2024). In addition, the Saharan climate is characterized by frequent sandy winds and significant dust accumulation, which further reduce the efficiency and performance of PV systems. Dust particles settle on the panel surface, obstructing sunlight and increasing maintenance requirements. (Wan et al., 2024).

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Effective cooling strategies are therefore critical to address these challenges and improve the performance and lifespan of PV systems. Cooling techniques for PV modules can be broadly categorized into active and passive methods (Kumar et al., 2024; Osman et al., 2024; Sharaf et al., 2022). Active cooling systems, such as forced air or water circulation, and thermoelectric coolers, require external energy inputs, while passive cooling methods, including natural airflow, heat sinks, or phase change materials (PCMs), rely on natural processes without additional energy consumption (Abou Akrouch et al., 2023, Gharzi et al., 2020, Al Miaari and Ali, 2023, Liu et al., 2023, Ranawade & Nalwa, 2023).

Recent research has explored various cooling techniques to improve PV performance. For example, Elminshawy et al. (2022) demonstrated the effectiveness of aluminum-finned heat sinks in reducing PV module temperatures by up to 10°C, which resulted in a 2.74% increase in electrical efficiency. Similarly, studies by Elbreki et al. (2021) investigated novel fin designs, which led to a 10.69% improvement in efficiency. Moreover, the use of PCMs to enhance thermal management has been explored by Sudhakar et al. (2021), who combined PCMs with natural water circulation to significantly improve PV performance. Although these studies provide valuable insights, most focus on passive cooling or hybrid approaches and do not address the specific challenges of extreme high-temperature environments such as the Saharan climate. Active cooling systems have been shown to effectively mitigate these temperature-related challenges. Yildirim et al. (2022) reported a 19.56% increase in electrical efficiency for water-cooled PV systems, demonstrating substantial improvements over conventional PV setups. Similarly, Shahverdian et al. (2021) evaluated the performance of a water flow cooling system applied to polycrystalline and monocrystalline PV modules. Their results showed significant improvements in power output and efficiency, with increases of up to 13.7 W and 14.8 W in power and 3.8% and 4.2% in efficiency for polycrystalline and monocrystalline modules, respectively. Khosravi et al. (2023) optimized water-cooled PV-thermal modules, improving both thermal performance and power generation. However, these studies primarily focus on controlled environments, and their findings may not be fully applicable to the real-world challenges faced in the Saharan context. Further advancements in cooling technologies include passive refrigeration systems. Chandrasekar et al. (2023) proposed a novel passive refrigeration technique for PV panels using a Capillary Pumped Loop (CPL) system with refrigerants R134a and R22. Their results showed a reduction in PV panel temperature by 2°C with R134a and 3°C with R22, leading to a 7.33% and 10.54% increase in electrical power, respectively. These findings underscore the potential of refrigeration-based cooling systems to enhance PV efficiency in extreme conditions. Additionally, studies have explored the integration of heat pumps with water-cooled PV systems. Zhang et al. (2025) introduced an innovative solar-air dual-source heat pump system coupled with a water-cooled photovoltaic/thermal system. Their experimental results demonstrated a significant boost in system efficiency, with electrical efficiency reaching up to 18.33% and collector efficiency reaching 98.94%. The study also showed that the heat pump's coefficient of performance could reach up to 4.2 under cloudy conditions, offering substantial operational and economic benefits. In another recent study, Navaei et al. (2024) investigated the effects of water spray cooling on PV panel performance. Their findings revealed that water spray not only reduces panel temperature but also cleans the surface, improving the long-term performance of the modules. Optimal nozzle configurations led to an efficiency enhancement of up to 22.72%, emphasizing the effectiveness of spray cooling for improving PV performance. Furthermore, Alsayegh et al. (2024) conducted an experimental investigation comparing three PV cooling systems: passive air cooling (FPA), active water cooling on the PV frontside (FAW), and active water cooling on the PV backside (BAW). Their findings revealed that the FPA, FAW, and BAW systems reduced PV surface temperatures by 2.1°C, 22.4°C, and 17.8°C, respectively. The electrical efficiency improved by 0.84% and 6.86% for the FPA and BAW systems, while the FAW system decreased efficiency by 26.27% due to significant light transmission losses caused by plexiglass and water layers. However, the FAW system exhibited higher thermal energy collection compared to the BAW system. These results indicate that the choice of cooling system should depend on whether the goal is to maximize electrical power output (favoring BAW) or thermal energy collection (favoring FAW).

This study seeks to address the gaps in current research by evaluating a novel active cooling system that directly flows water over the surface of PV panels, designed specifically for the Saharan climate. The cooling system was tested at different flow rates (0.17 and 0.037 L/s) and activation temperatures (35°C and 45°C) to assess its impact on both electrical efficiency and thermal performance. Unlike previous studies, this work provides a comprehensive experimental analysis of the system's performance under real-world conditions, while also considering the potential for water reuse. This study aims to offer valuable insights for optimizing PV system performance in extreme climates and contribute to the development of effective cooling strategies for solar energy generation in hot regions.

The experimental study was carried out at the University of Ouargla (latitude: 31.95°N, longitude: 5.33°E), Algeria, from May 15 to June 1, 2022, during clear sunny days to reflect the typical Saharan climatic conditions.

Materials and methods

Experimental setup

The system applied water directly to the PV panel when it exceeded a temperature threshold, preventing overheating and improving module efficiency and longevity. Ouargla, known for its high solar irradiation and long sunny days, makes it an ideal location for photovoltaic power generation (Chaich et al., 2024). However, it also experiences some of the highest ambient temperatures, often exceeding 43°C (Gougui et al., 2020), which causes overheating and reduces power generation efficiency.

The experimental apparatus was installed on top of the University of Ouargla building (latitude: 31.95°N). The test rig consisted of two configurations: a water-cooled PV system and a standard PV panel for direct comparison. The PV modules' properties are listed in Table 1. Systems were mounted at a 30° tilt angle, optimal for the region (Dida et al., 2020), on white-sprayed metal brackets to minimize heat conduction and prevent oxidation. Mounting dimensions (56 cm × 85 cm) allowed adjustments in the tilt angle. The panels were elevated 80 cm above ground to reduce heat exchange.

The cooling system operates intermittently, activating only when the PV panel reaches predefined thresholds of 35°C or 45°C, significantly reducing water consumption and optimizing resource efficiency. The water flow (Q_v) is designed to avoid negatively impacting the PV panel's optical properties, such as transmittance, reflectivity, and emissivity, due to its thin and evenly distributed water layer. The system ensures uniform water distribution using evenly spaced holes, though further research is recommended to assess the impact of water flow characteristics on optical and thermal performance.

Table 1 Datasheet for polycrystalline PV

Module type	50 P (36)
Maximum power	50 W
Maximum voltage (V_{max})	17.98 V
Maximum courant (I_{max})	2.7 A
Voltage in open circuit (V_{oc})	21.87 V
Courant in a close circuit (I_{cc})	3.04 A
Dimensions	830 × 540 × 28 (mm)
Weight	6 (Kg)
Tolerance	3%
System operating voltage	1000 V
Standard test condition, AM1	1000W/m2, 5.25°C
Cells	36Pcs,156 × 63 polycrystalline Silicon

In the Saharan region, where summer ambient temperatures often exceed 25°C and frequently surpass 30°C (Necib & Necib, 2020), the water available for cooling naturally has a temperature above 25°C. During the experiment, the water temperature consistently remained around 28°C. Lowering the water temperature to 25°C would require substantial energy input, rendering the process impractical. Consequently, threshold temperatures of 35°C and 45°C were selected to align with the regional climatic conditions, ensuring effective cooling while minimizing energy consumption.

Recovered water was collected in a 50 mm PVC pipe for reuse or storage. If water exceeded 30°C, it was stored to cool before irrigation. The total mass of water used for cooling is calculated in Table 2 and discussed in detail in the results section.

The cooling system included an AC water pump, 25 mm PVC tubing, and seven evenly spaced holes for water distribution. Water flow was activated at set temperatures of 35°C and 45°C and tested at flow rates of 0.037 L/s and 0.17 L/s. Figure 1 illustrates the experimental setup. However, it is important to note that the use of a pump is not strictly necessary. If water flow can be provided by a gravitational setup or a pressurized water supply network, the operational cost of the system can be significantly reduced, making it more economically viable for large-scale applications.

Measurement and data collection

Temperature measurements

K-type thermocouples were installed on the rear faces of both the cooled and reference (uncooled) PV panels, as well as on the front face of the reference panel, to monitor temperature differences. The negligible difference between front and rear temperatures (2.5%) allowed for a focus on rear temperature measurements, which were recorded consistently every 20 min from 8:20 a.m. to 4:00 p.m.

Electrical performance

Voltage and current measurements were performed using digital multimeters to calculate the electrical power output and efficiency. These metrics were critical in evaluating the cooling system's impact on PV performance.

Environmental monitoring

Solar irradiance was measured using a Digital Solarimeter 1307SI, while ambient temperature was monitored with thermocouples positioned near the experimental setup. These environmental measurements ensured accurate correlation with the observed PV performance.

Table 2 Impacts on Heat Transfer and Electrical Efficiency

Module type	Flow rate of 0.17 L/s and set panel temperature of 35°C	Flow rate of 0.037 L/s and set panel temperature of 35°C	Flow rate of 0.17 L/s and set panel temperature of 45°C	Flow rate of 0.037 L/s and set panel temperature of 45°C
Average temperature of water in (°C)	28.12	29.1	28.1	25.9
Average temperature of water out (°C)	28.95	32.67	28.52	38.7
Max temperature reduction (°C)	30.7	28.7	33	31.1
Average temperature reduction (°C)	22.25	21.08	24.69	23.83
Total mass of water used for cooling (kg)	4692.17	1021.237	3672.17	399.637
Total time of cooling (s)	27,601	27,601	21,601	10,801
Panel area (m^2)	0.4482	0.4482	0.4482	0.4482
volume flow rate (m^3/s)	0.00017	0.000037	0.00017	0.000037
Average heat energy transferred to water (MJ/m^2 of PV)	36.153	34.018	14.433	9.825
Average Electrical efficiencies for Panel with cooling/without cooling system (%)	18.44/15.31	17.59/14.64	18.26/15.52	16.97/14.36
Average electrical efficiency improvement(%)	20.55	20.15	17.65	18.18

Formula for power calculation

To calculate the total input power to the PV system, the following formula was employed:

$$P_{inlet} = I_R \times S \tag{1}$$

where S is the PV surface area and I_R is the solar radiation measured at the PV surface. While the maximum electrical output power was determined by:

$$P_{\max} = V_{optimal} \times I_{optimal} \tag{2}$$

where (P_{max}) denotes maximum power, $I_{optimal}$ and $V_{optimal}$ denotes optimum current and voltage, respectively. The PV electrical efficiency η_e was calculated as follows:

$$\eta_e = \frac{P_{\max}}{P_{inlet}} \times 100 \tag{3}$$

Results and discussion

The experiments controlled the water flow rate and calculated PV efficiency improvements to determine the optimal flow rate and activation temperature for the Ouargla region.

Solar radiation and ambient temperature variation

Figure 2 illustrates the evolution of solar radiation and ambient temperature recorded from 08:20 to 16:00 during the experimental days (May 15, 29, 31, and June 01). These days exhibited nearly identical variations in ambient temperature and solar radiation, with the exception of May 29, when the weather briefly became cloudy. The minimum and maximum average solar radiation during the experimental period ranged from 302.67 to 940.75 W/m^2. The average solar radiation values for May 15, 29, 31, and June 01 were 794.54, 697.04, 738.54, and 733.83 W/m^2, respectively, with the lower value on May 29 attributed to partly cloudy weather (see Fig. 2a).

As shown in Fig. 2b, ambient temperature exhibited minor variations, with maximum temperatures recorded as 42.8°C, 41.9°C, 44.8°C, and 45.6°C on May 15, 29, 31, and June 01, respectively. Under these conditions, the temperature of the uncooled PV panel ranged from 30 to 70°C, with an average panel temperature of 68°C. This high temperature significantly impacts the electrical energy output of the PV panel, emphasizing the importance of effective cooling strategies.

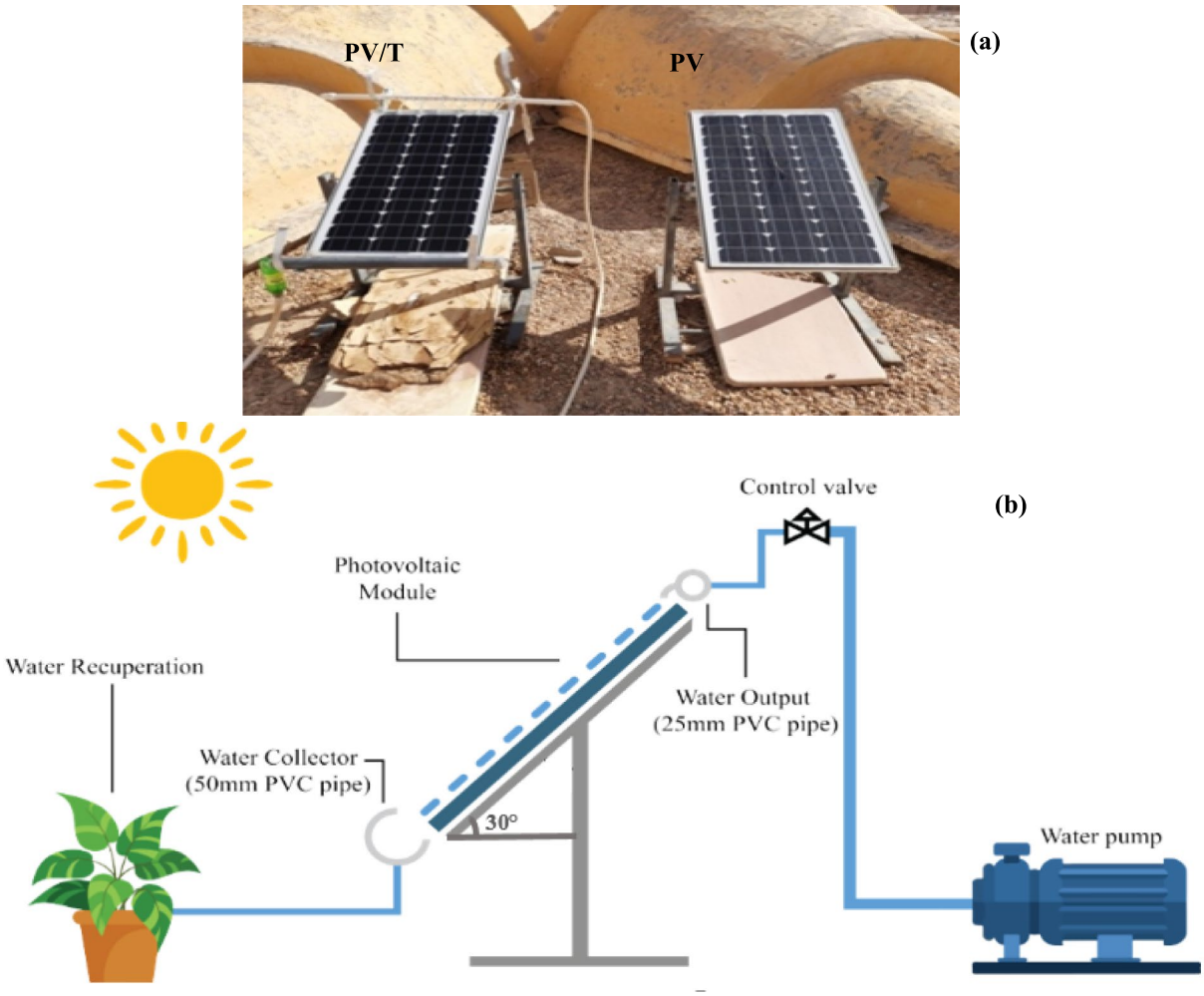


Fig. 1 **a** Experimental test setup front view; **b** Schematic diagram of an experimental process of the cooling system

Temperature reduction

Maximum flow rate (0.17 L/s) and set panel temperature of 35°C

Figure 3 presents the experimental analysis of temperature reduction in a photovoltaic (PV) module with cooling compared to a module without cooling at a mass flow rate of 0.17 L/s and a set panel temperature of 35°C. As shown in Fig. 3a, the maximum temperature difference between the cooled and uncooled PV modules reached 30.7°C at 12:00 p.m., demonstrating the substantial cooling capacity of the system. The average temperature reduction provided by the cooling system across all time intervals was 22.25°C, highlighting its consistent performance in reducing module temperatures.

Additionally, the percentage reduction in temperature ranged from 19.75 to 47.60% throughout the day (see Fig. 3b), indicating that the cooling system's effectiveness varies depending on the time of day and environmental conditions.

Minimum flow rate (0.037 L/s) and set panel temperature of 35°C

Figure 4 illustrates the temperature reduction in the PV module with cooling at a flow rate of 0.037 L/s and a set panel temperature of 35°C, compared to the uncooled module. The maximum temperature difference between the cooled and uncooled PV modules was 28.7°C, observed at 13:00. The average temperature reduction provided by the cooling system across all time intervals was 21.08°C (see Fig. 4a).

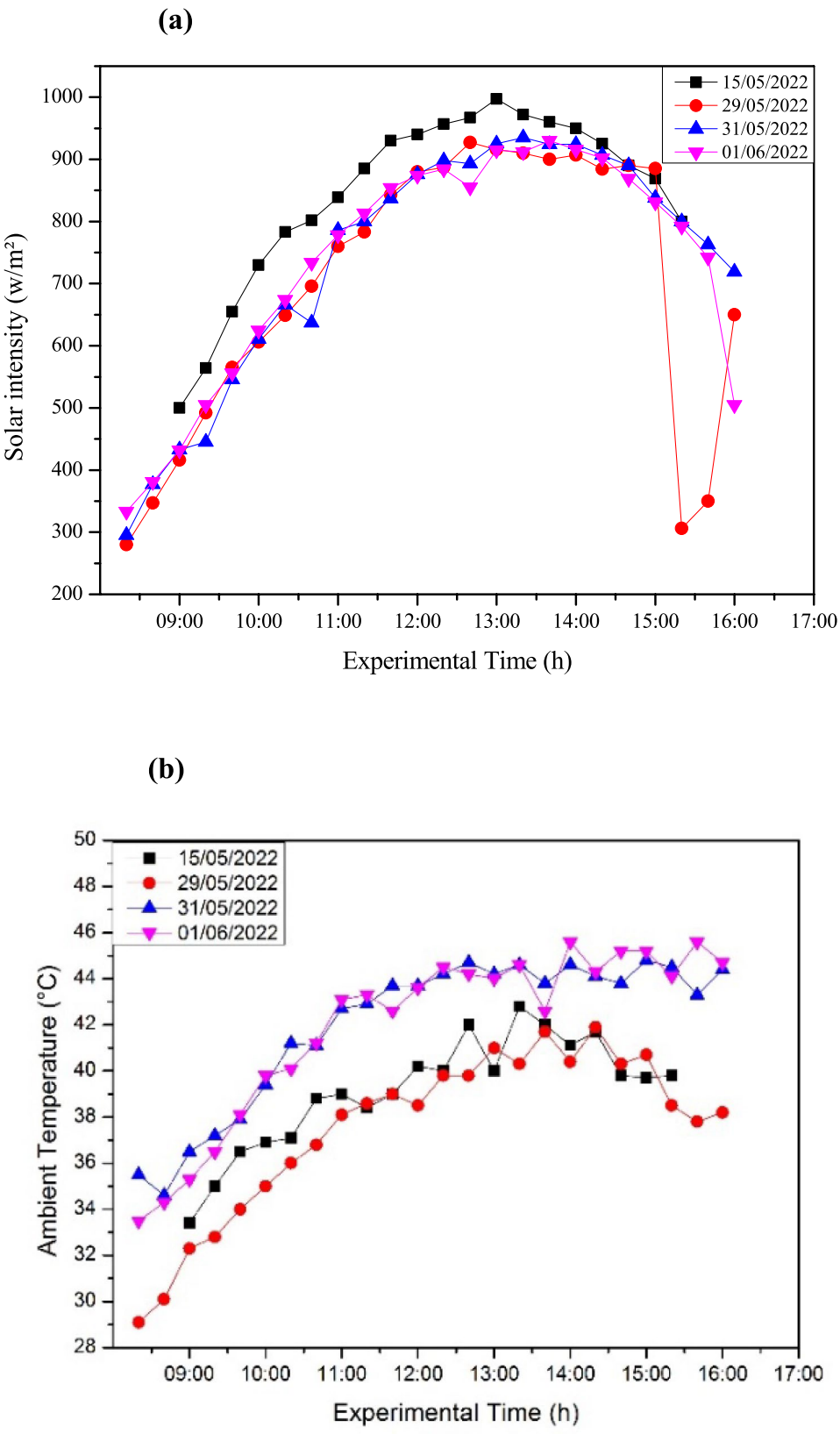


Fig. 2 The variation of: **a** solar radiation, **b** Ambient temperature

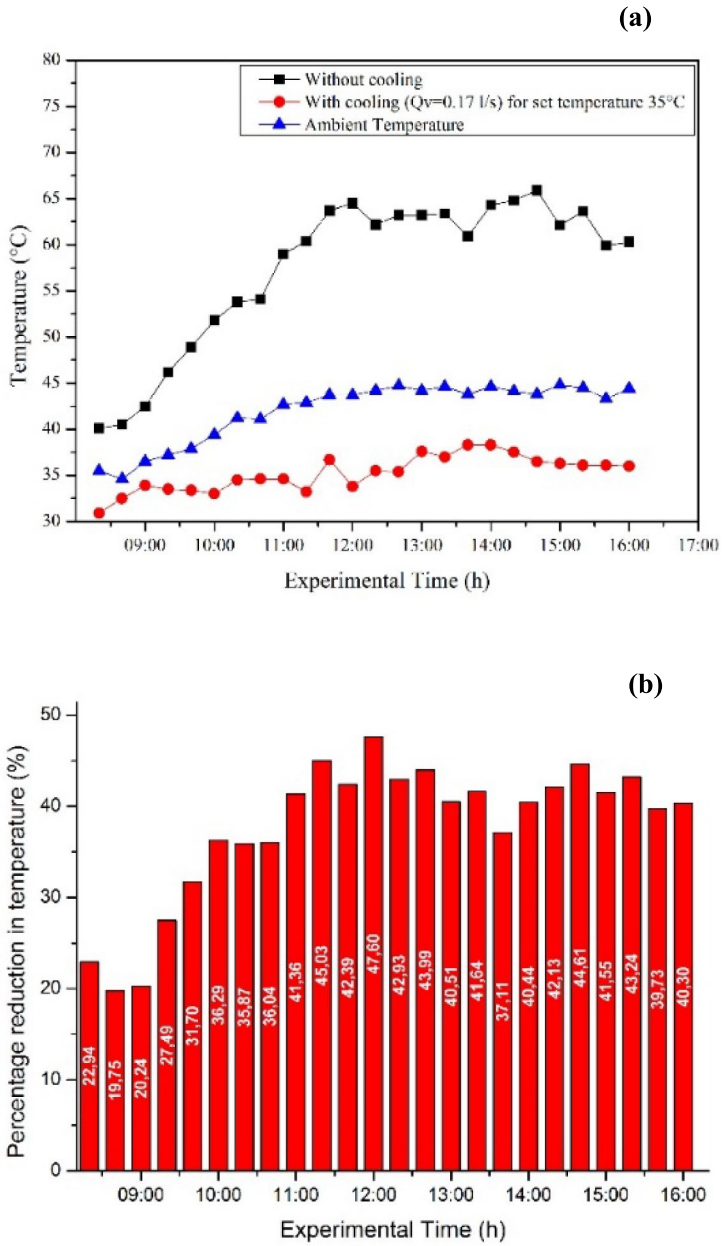


Fig. 3 Experimental analysis of: **a** temperature variation, **b** temperature reduction, in photovoltaic (PV) module with and without cooling at a mass flow rate of 0.17 L/s and set panel temperature of 35°C

The percentage temperature reduction achieved by the cooling system at each time interval is shown in Fig. 4b. These percentages ranged from 0.58 to 43.03%. The minimum relative difference in temperature (0.58%) was observed at 8:20 a.m., attributed to the panel temperature being below the activation threshold of 35°C, preventing the cooling system from operating. The average percentage temperature reduction achieved by the cooling system over the entire experimental period was 37.98%, indicating a significant cooling effect provided by the system.

Effect of flow rate on PV temperature reduction (PV cooling at 35°C)

Comparing the percentage reduction in temperature provided by the cooling system at two different mass flow

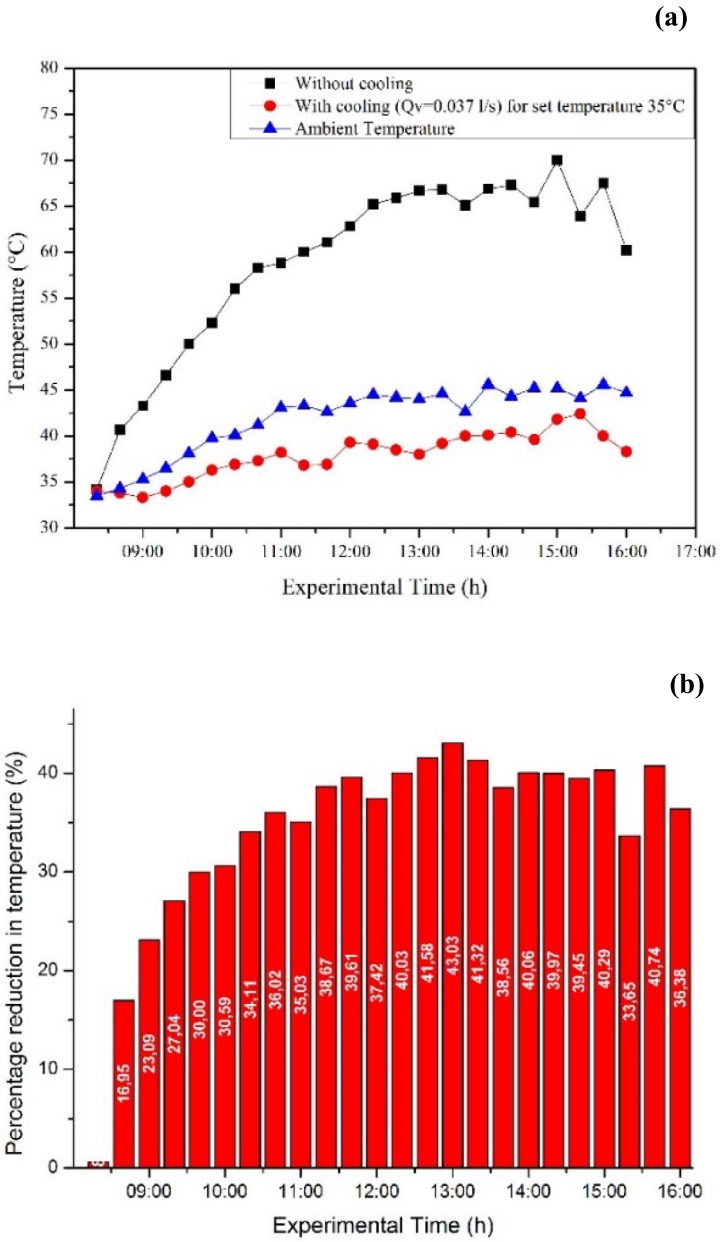


Fig. 4 Experimental analysis of: **a** temperature variation, **b** temperature reduction, in photovoltaic (PV) module with and without cooling at a mass flow rate of 0.037 L/s and set panel temperature of 35°C

rates (0.17 L/s and 0.037 L/s) for a set panel temperature of 35°C (Fig. 5), it can be observed that the higher mass flow rate of 0.17 L/s generally results in a greater percentage reduction in temperature compared to the lower mass flow rate of 0.037 L/s. For the maximum flow rate, the median percentage reduction in temperature achieved by the cooling system is 40.47%, whereas it is 37.98% for the minimum flow rate.

Maximum flow rate (0.17 L/s) and set panel temperature of 45°C

The experimental analysis of temperature reduction in a PV module with and without cooling at a mass flow rate of 0.17 L/s and a set panel temperature of 45°C is shown in Fig. 6. At 9:20 a.m., the temperature of the PV panel exceeds the threshold of 45°C, triggering the activation of the cooling

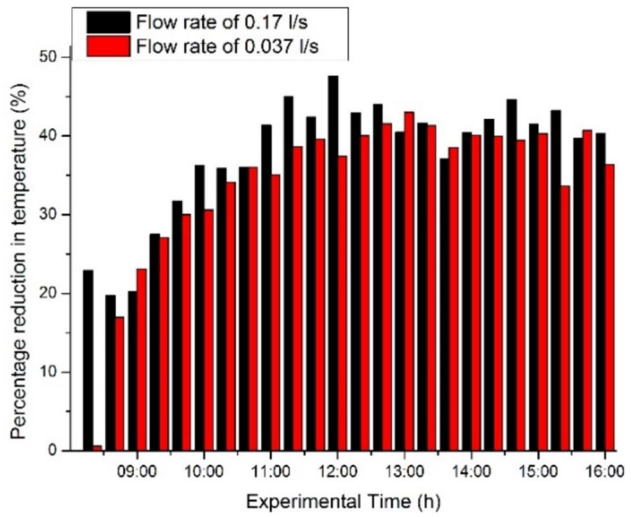


Fig. 5 Effect of flow rate in reduction of PV temperature module for set panel temperature of 35°C

system (Fig. 6a). As the ambient temperature increases throughout the day, the cooling system effectively reduces the temperature of the PV module.

The maximum temperature difference of 33°C, observed at 1:40 p.m., highlights the system's capability to mitigate heat buildup and maintain lower module temperatures. Additionally, the average temperature reduction of 24.69°C demonstrates the system's consistent performance in regulating the PV module's temperature throughout the day.

Minimum flow rate (0.037 L/s) and set panel temperature of 45°C

Figure 7 shows the variation in the relative temperature difference between photovoltaic (PV) modules with and without cooling at a set panel temperature of 45°C and a mass flow rate of 0.037 L/s. The results indicate a maximum temperature difference of 31.1°C (47.4%), observed at 11:40 a.m., highlighting the cooling system's effectiveness in reducing the PV module temperature. This significant temperature difference demonstrates the system's ability to dissipate heat efficiently, preventing the PV modules from reaching excessively high temperatures compared to the uncooled condition.

The activation of the cooling system at 9:00 a.m. marks the threshold temperature being reached, triggering the cooling mechanism. This timely activation is crucial to maintaining the PV modules within the desired temperature range. The average temperature reduction achieved by the cooling system across all time intervals is 23.83°C, reflecting the system's consistent cooling performance throughout the observation period.

Effect of flow rate on PV temperature reduction (cooling at 45°C)

By comparing the percentage reduction in temperature of PV for two different flow rates (0.037 L/s and 0.17 L/s) at the same time intervals and for set panel temperature of 45°C (Fig. 8), we can see that the flow rate of 0.17 L/s generally results in higher percentage reductions in temperature compared to the flow rate of 0.037 L/s. This indicates that a higher flow rate leads to more effective cooling of the PV module. It is important to note that specific temperature reduction percentages are influenced by various factors such as ambient conditions, cooling duration, and flow rate. Further analysis and comparison with additional data would be required to draw more definitive conclusions regarding the performance of cooling systems at different flow rates. Overall, the data suggests that a higher flow rate of 0.17 L/s tends to provide greater percentage reductions in temperature for the PV module compared to a flow rate of 0.037 L/s. This information can be valuable in optimizing the cooling system design and operation to achieve the desired temperature reduction and improve the performance of the PV module.

Electrical efficiency improvement

The short-circuit current and the open-circuit voltage

Figure 9 illustrates the variations in short-circuit current (I_{sc}) for the tested PV modules under two conditions: without cooling and with cooling, at a set panel temperature of 35°C and a mass flow rate of 0.17 L/s. The short-circuit current is a critical parameter that directly influences the performance and efficiency of PV modules. Without cooling, the short-circuit current ranges from 1.45 A at 9:00 a.m. to 2.53 A at 4:00 p.m. Conversely, with cooling, the short-circuit current values range from 1.58 A at 9:00 a.m. to 2.6 A at 4:00 p.m.

The relative difference in short-circuit current between cooled and uncooled PV modules varies throughout the day. The maximum, minimum, and average relative differences are 9.68%, 2.27%, and 4.33%, respectively, highlighting the cooling system's impact on improving short-circuit current and overall panel performance.

Figure 10 demonstrates that the application of the cooling system also increases the open-circuit voltage of the PV panel compared to the uncooled panel, under the same flow rate and set panel temperature. The relative difference in open-circuit voltage ranges from 10.75 to 19.09%, with an average improvement of 16.16%. These findings underscore the significant positive influence of the cooling system on both short-circuit current and open-circuit voltage, resulting in enhanced PV panel performance and efficiency. Furthermore, as shown in Fig. 11, the output power of the cooled panel achieves a maximum enhancement rate of 24.68%

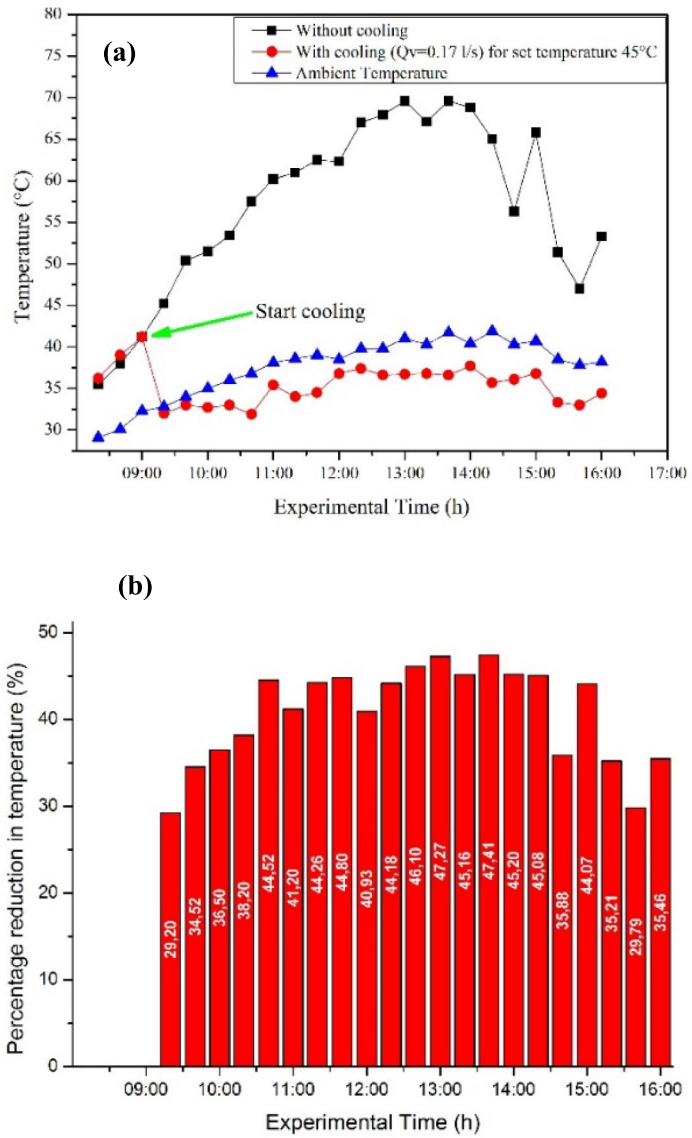


Fig. 6 Experimental analysis of: **a** temperature variation, **b** temperature reduction, in photovoltaic (PV) module with and without cooling at a mass flow rate of 0.17 L/s and set panel temperature of 45°C

during the experimental period (8:20 a.m. to 4:00 p.m.) and an average improvement of 20.55%. This underscores the importance of implementing a cooling system to achieve higher efficiency and overall better performance of PV modules.

Maximum power point (MPP)

Figure 12 presents the results of an experiment investigating the effect of cooling on the performance of a photovoltaic (PV) system. The experiment was conducted at two set panel temperatures, 35°C and 45°C, with two cooling water flow rates: 0.17 L/s and 0.037 L/s.

The results demonstrate that cooling significantly enhances the performance of the PV system. At both set temperatures, the PV system with cooling achieved a higher maximum power point (MPP) and system efficiency compared to the PV system without cooling. Additionally, the flow rate of the cooling water influenced the performance. At both set temperatures, the PV system with a higher cooling water flow rate (0.17 L/s) exhibited a

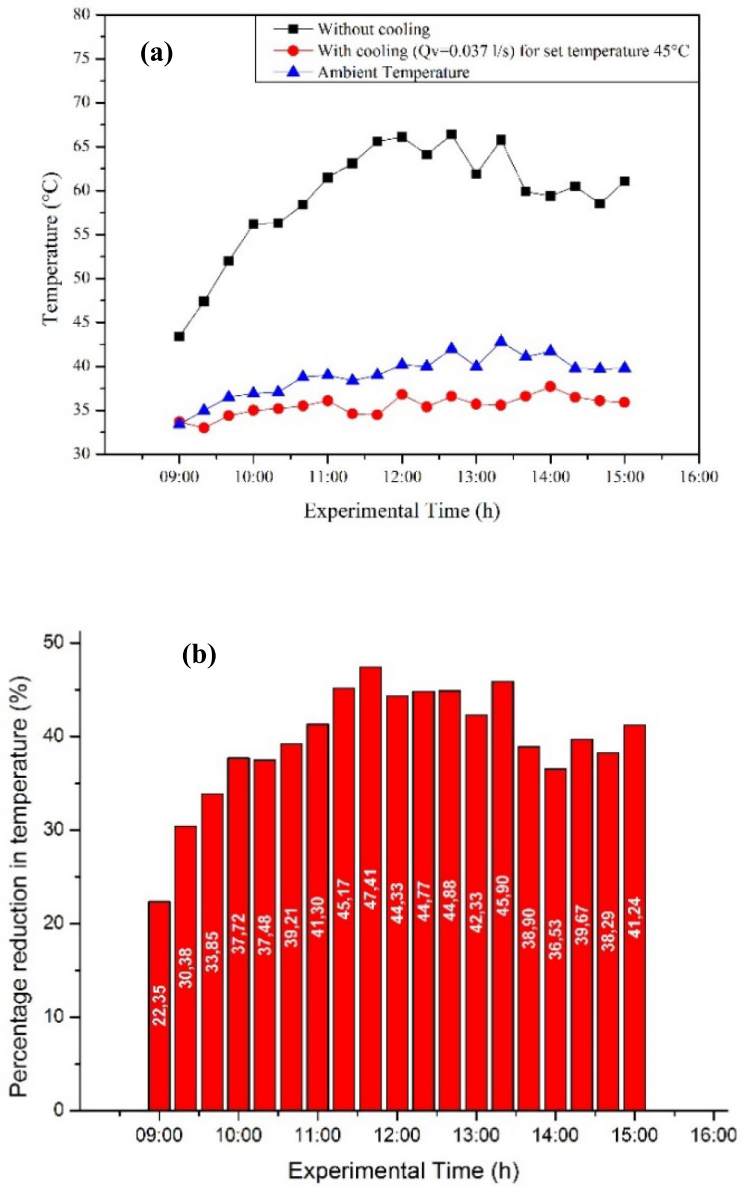


Fig. 7 Experimental analysis of: **a** temperature variation, **b** temperature reduction, in photovoltaic (PV) module with and without cooling at a mass flow rate of 0.037 L/s and set panel temperature of 45°C

greater MPP and system efficiency than the system with a lower flow rate (0.037 L/s).

The improvements were more pronounced at the higher set temperature (45°C). At this temperature and the higher flow rate of 0.17 L/s, the PV system with cooling achieved an MPP of 51.832 W, which is 30.50% higher than the MPP of the PV system without cooling (39.7194 W). Similarly, the system efficiency of the cooled PV system was 12.85%, representing a 30.50% improvement compared to the efficiency of 9.85% for the uncooled system.

Electrical efficiency

Figure 13 illustrates the electrical efficiency of photovoltaic (PV) panels with and without cooling. The experiment was conducted over a period of 7 h, from 9:20 a.m. to 4:00 p.m., with PV panels cooled using water at a flow rate of 0.17 L/s. Two set panel temperatures, 35°C and 45°C, were tested to regulate the activation of cooling. The results indicate that cooling significantly improves the electrical efficiency of PV panels. The maximum improvement in

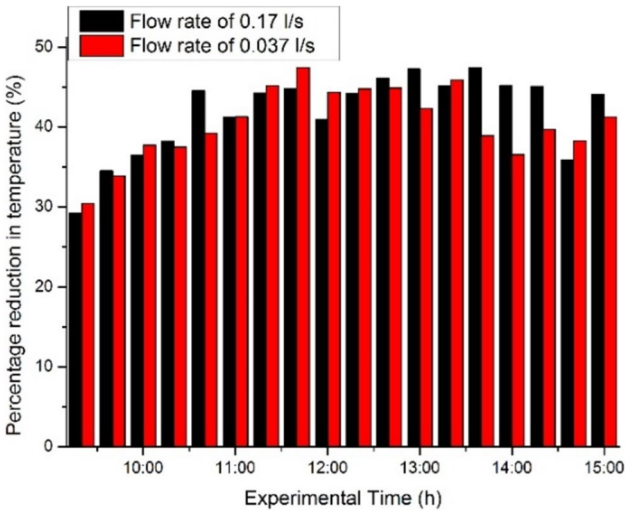


Fig. 8 Effect of flow rate in reduction of PV temperature module for set panel temperature of 45°C

electrical efficiency was 24.68% for cooling with water at 35°C and 23.20% for cooling at 45°C. This improvement was most pronounced during the hottest part of the day, between 10:00 a.m. and 2:00 p.m. In addition to enhancing electrical efficiency, cooling PV panels helps extend their lifespan. By reducing the operating temperature, cooling minimizes the rate of degradation of the PV cells, leading to increased longevity and a higher return on investment.

Electrical power improvement

Figure 14 illustrates the electrical power improvement of cooled PV panels at different set temperatures and flow rates, focusing on short-circuit current and open-circuit voltage. The experiment was conducted with flow rates of 0.17 L/s and 0.037 L/s, and cooling system activation temperatures of 35°C and 45°C. The results demonstrate that cooling significantly enhances the electrical power output of PV panels. The maximum electrical power improvement observed was 24.68% with a flow rate of 0.17 L/s at a set temperature of 35°C. The improvement in electrical power output was more pronounced at higher flow rates and lower set temperatures. Additionally, the flow rate of the cooling water had a notable impact on electrical power enhancement. Increasing the flow rate from 0.037 to 0.17 L/s resulted in power improvements of up to 2.70% at a set temperature of 35°C and 2.76% at 45°C. These findings highlight the importance of optimizing both flow rate and activation temperature to maximize the cooling system's effectiveness in improving PV panel performance.

Optimal PV cooling configuration

Optimizing photovoltaic (PV) cooling is a critical aspect of achieving efficient solar energy conversion. The right cooling configuration significantly impacts PV panel performance, energy output, and system longevity. Balancing cooling techniques, including flow rates and set panel

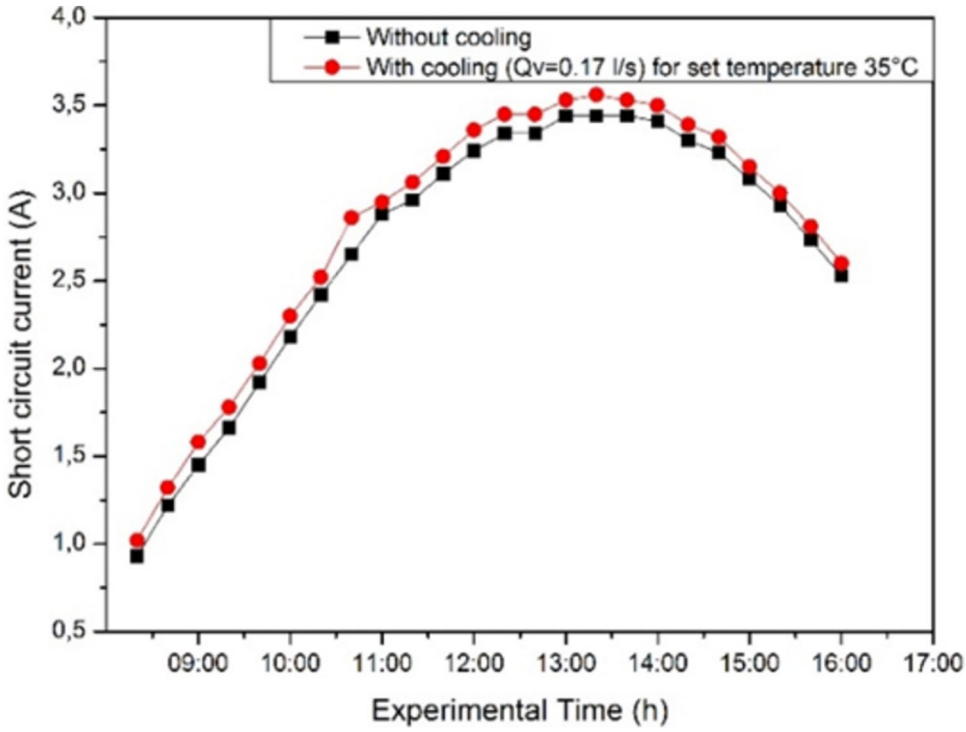


Fig. 9 Variation of short-circuit current with and without cooling for set temperature 35°C and mass flow rate of 0.17 L/s

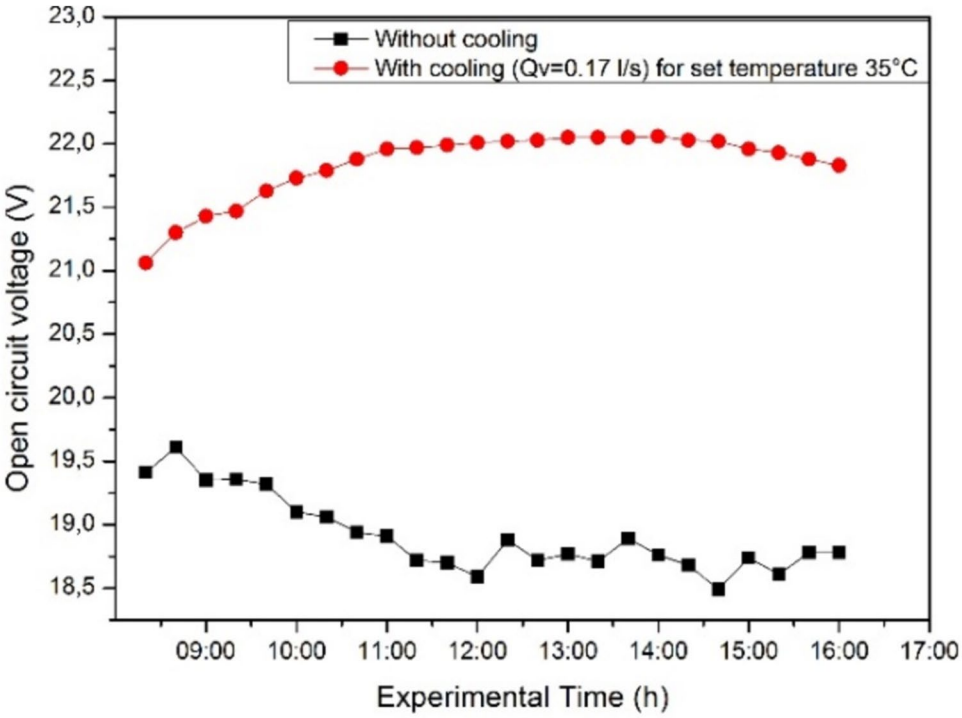


Fig. 10 Variation of open circuit voltage with and without cooling for set temperature 35°C and mass flow rate of 0.17 L/s

temperatures, is essential to maximizing the benefits of PV technology while addressing resource limitations.

Table 2 provides data on the amount of heat transferred to the water under various cooling conditions, focusing on flow rates and set panel temperatures. The results highlight that the optimal case depends on the primary objective, whether it is cooling the PV module or utilizing waste heat for water heating. Higher flow rates generally result in better cooling of the PV module, while higher set panel temperatures enhance heat transfer to the water.

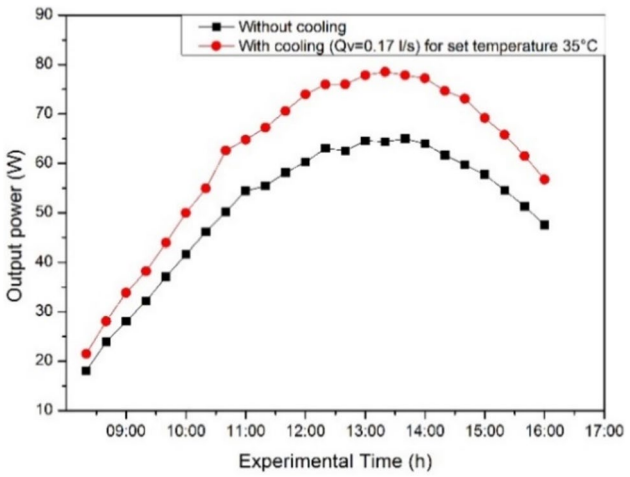


Fig. 11 Variation of power output with and without cooling for set temperature 35°C and mass flow rate of 0.17 L/s

If the primary goal is to cool the PV module and prevent overheating, a higher flow rate (0.17 L/s) and a lower set panel temperature (35°C) are more effective. This combination yields a higher average heat transfer to the water (36.153 MJ/m^2 of PV), ensuring efficient cooling. Conversely, if the objective is to utilize waste heat for water heating, a lower flow rate (0.037 L/s) and a higher set panel temperature (45°C) are preferable. This configuration results in an average heat transfer of 9.825 MJ/m^2 of PV, demonstrating reduced dissipation and increased heat utilization.

Another key aspect is the total mass of water used for cooling. Higher flow rates lead to greater water consumption, making it critical to balance cooling efficiency and water resource sustainability, particularly in regions with scarce water availability. Furthermore, the total cooling time varies with flow rate, with lower flow rates requiring longer durations.

The table also illustrates the average heat energy transferred to the water per square meter of PV module under different conditions, showcasing the variability in heat transfer based on cooling settings. The optimal case depends on the specific objective, whether maximizing or minimizing heat transfer. Additionally, the average electrical efficiencies for panels with and without cooling systems are provided. These values demonstrate consistent improvements in electrical efficiency due to the cooling system across all cases.

While higher flow rates enhance cooling performance, their practicality may be limited by water resource constraints, especially in arid regions. This underscores the

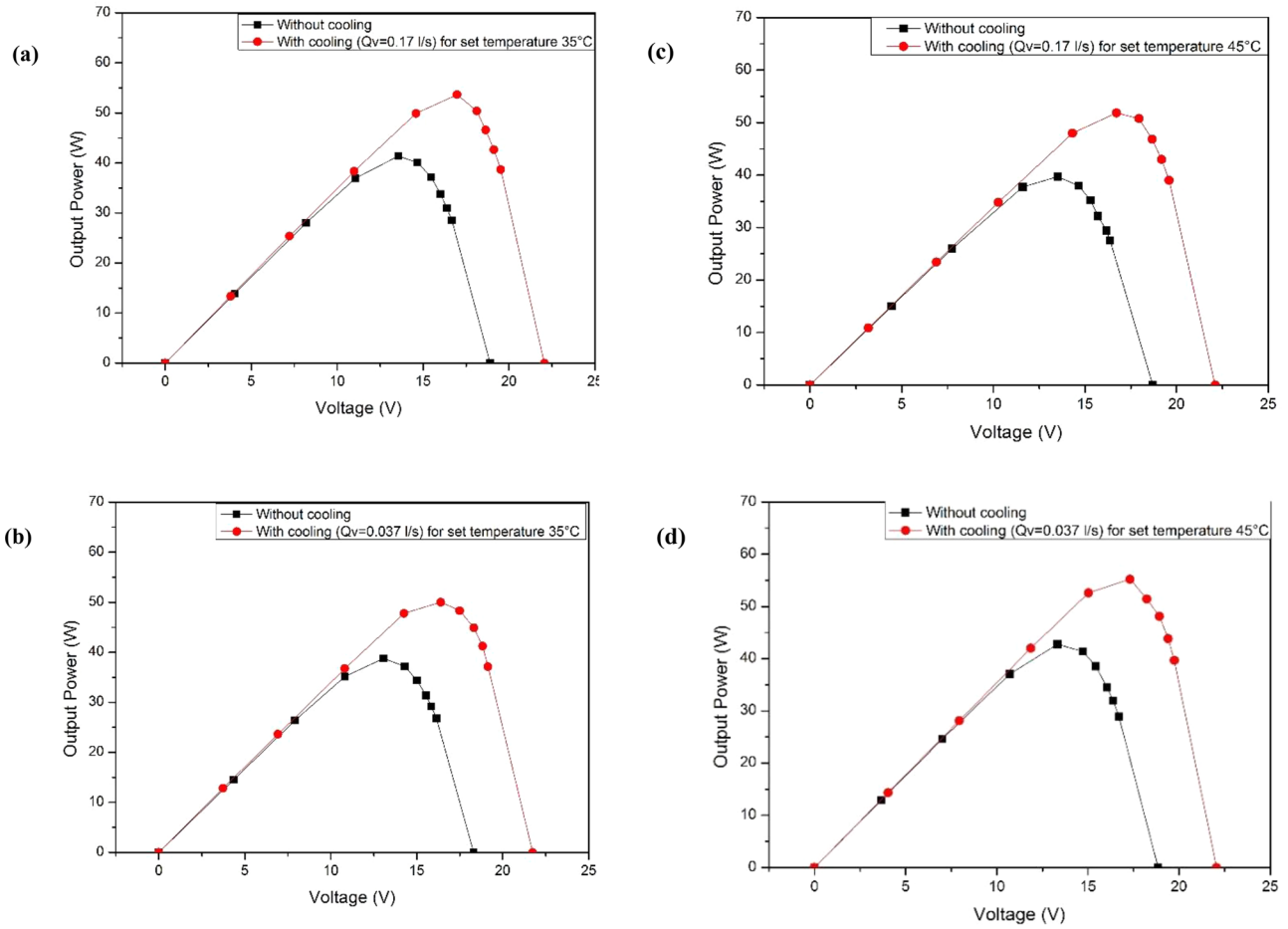


Fig. 12 Output power as a function of voltage variation for: **a** Qv = 0.17 L/s and set temperature of panel 35°C, **b** Qv = 0.037 L/s and set temperature of panel 35°C, **c** Qv = 0.17 L/s and set temperature of panel 45°C, **d** Qv = 0.037 L/s and set temperature of panel 45°C

importance of optimizing flow rates to balance cooling efficiency with sustainable water usage.

In conclusion, the analysis emphasizes the importance of evaluating multiple factors when selecting the optimal cooling configuration. The amount of heat gained by the water, the improvement in electrical efficiency, and the mass of water consumed are all critical considerations. Depending on water availability and the desired objective (cooling the PV or heating the water), the optimal configuration can vary. A comprehensive evaluation of these parameters is essential for making informed decisions on PV cooling strategies.

Comparison of present results with previous

The comparative analysis highlights the strengths and weaknesses of various cooling methods for photovoltaic (PV) systems. The present study's active water cooling approach demonstrates a significant temperature reduction (up to 33.0 °C) and electrical efficiency improvement (mean: 20.55%, max: 24.68%) under real-world Saharan climate conditions. Additionally, our system is simple to install and does not require expert intervention, making it more accessible for widespread use. It is also highly effective in cleaning PV panels, which is particularly advantageous in climates characterized by sandy winds and dust accumulation. Moreover, the system has the potential to utilize recovered thermal energy for additional applications, further enhancing its utility. These results outperform passive systems like those studied by Sudhakar et al. (2021), where the average temperature reduction was limited to 5.4 °C with a 13.54% efficiency increase. The combination of phase change material (PCM) with water cooling, while effective for thermal management, but may be more costly in high-temperature environments. Similarly, while Alsayegh et al. (2024) reported a higher temperature reduction for their FAW and BAW systems, their FAW system resulted in efficiency losses due to light transmission issues. Chandrasekar et al. (2023) proposed an innovative passive refrigeration system; however, it is less effective in reducing PV temperatures and improving electrical efficiency compared to our system. Additionally,

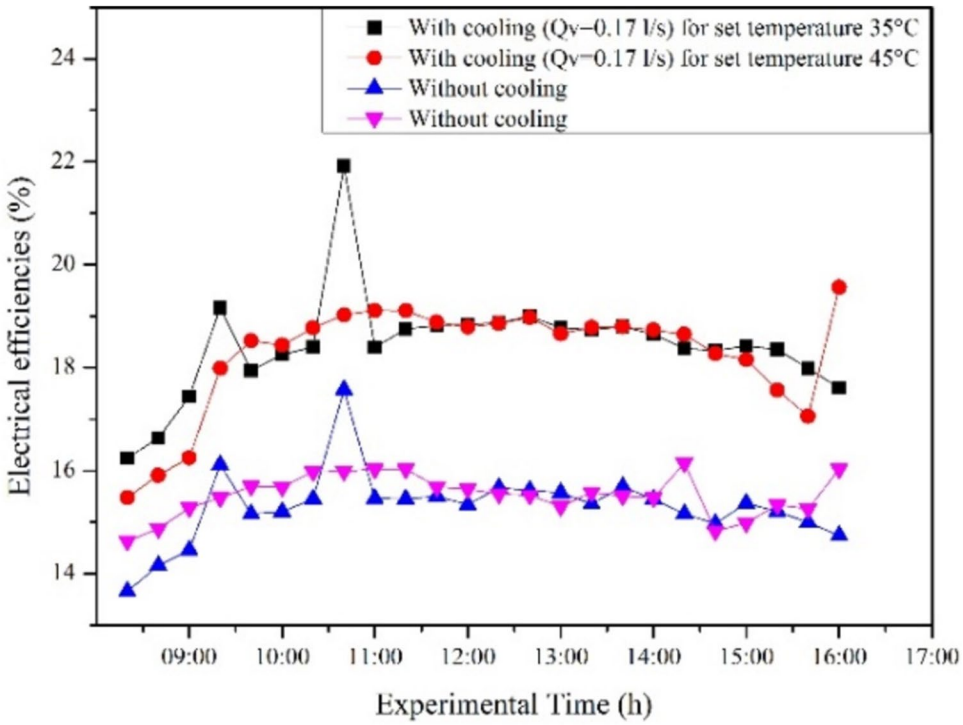


Fig. 13 Electrical efficiency comparison of PV from two different set temperatures, 35°C and 45°C and for maximum flow rates of cooling water, 0.17 L/s

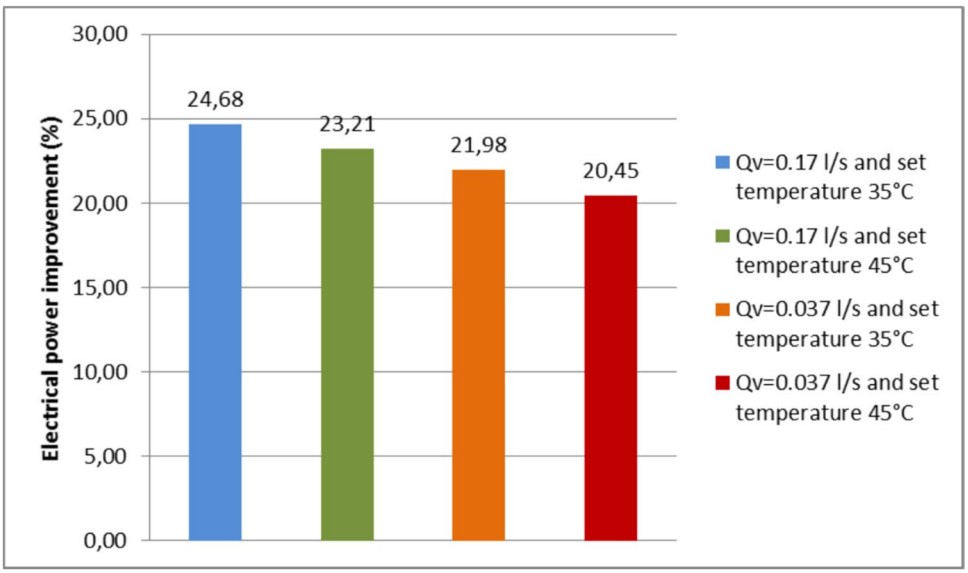


Fig. 14 Electrical power improvement of the cooled panels for different set temperatures and different flow rates

the use of refrigerants like R134a and R22 raises concerns about potential fire hazards and environmental pollution.

This study lies in tailoring the cooling system to high ambient temperatures and incorporating water reuse potential, addressing both energy efficiency and resource sustainability. Furthermore, the intermittency of water flow (activated only at specific temperatures) and its homogeneity across the panel surface ensure minimal impact on optical properties, such as transmittance and reflectivity, a key consideration absent in several other studies (Table 3).

Conclusion

This study provides a comprehensive evaluation of a water-based cooling system designed to enhance the performance of photovoltaic (PV) modules under the extreme climatic conditions of the Saharan region. The system employs direct water flow, activated at set temperatures of 35°C and 45°C, demonstrating significant improvements in PV performance by reducing module temperatures by up

Table 3 Comparison of Cooling Systems and Their Effects on PV Performance

Study	Cooling Method	Max Temperature Reduction (°C)	Electrical Efficiency Improvement (%)	Remarks
Present Study	Active water cooling (flow rates: 0.037 and 0.17 L/s, activation at 35 °C and 45 °C)	28.7–33.0	20.55 (mean), up to 24.68	Tailored for high ambient temperatures in Saharan climate with water reuse potential
Alsayegh et al. (2024)	FAW and BAW systems	22.4 (FAW), 17.8 (BAW)	– 26.27 (FAW), + 6.86 (BAW)	BAW improves efficiency, FAW impacts light transmission
Chandrasekar et al. (2023)	Passive refrigeration (CPL with R134a, R22)	2 (R134a), 3 (R22)	7.33 (R134a), 10.54 (R22)	Effective for controlled environments with refrigerants
Navaei et al. (2024)	Water spray cooling	0.35–0.7 per °C reduction	Up to 22.72	Cooling also aids cleaning; optimal nozzle configuration required
Sudhakar et al. (2021)	PCM with natural water circulation	5.4	13.54	Combines PCM with water cooling for better thermal management

to 33.0°C and enhancing electrical efficiency by as much as 24.68%.

The experimental results highlight the system's effectiveness in mitigating the overheating challenges faced by PV modules in high-temperature regions. On average, the cooling system reduced module temperatures by 21.08°C to 24.69°C, ensuring stable operation during peak solar irradiation hours. Additionally, the system achieved an average improvement of 20.55% in electrical efficiency, underscoring its ability to optimize energy conversion and maintain consistent performance under extreme climatic conditions.

Resource efficiency was another critical advantage of the system. By utilizing an intermittent water flow mechanism, the cooling process minimized water consumption while maintaining uniform and effective cooling across the PV surface. Importantly, the water flow did not compromise the optical properties of the PV panels, such as transmittance, reflectivity, and emissivity, ensuring sustained efficiency throughout the experimental period. Furthermore, the recovered water can be stored and reused for applications such as irrigation. This integration of water reuse not only enhances resource sustainability but also makes the system particularly valuable for arid regions where water is scarce.

Compared to existing cooling methods such as passive cooling and spray systems, the proposed solution offers several advantages. It's simple and cost-effective design ensures ease of installation and maintenance, without requiring expert intervention. Additionally, the potential for integrating thermal energy recovery adds further value, contributing to the system's overall efficiency and sustainability. These features make it a practical and scalable solution for optimizing solar energy systems in resource-limited environments.

Future research should focus on further optimizing water flow characteristics, such as thickness and distribution, to better understand their impact on the optical and thermal properties of PV panels. Additionally, exploring the system's adaptability to varying environmental conditions can enhance its applicability across diverse regions. By addressing these challenges, the findings of this study contribute to the development of effective and sustainable cooling strategies for PV modules, ensuring enhanced performance and longevity in regions facing extreme climate challenges.

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Data availability The datasets analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no financial or non-financial interests exist that are directly or indirectly related to this work.

Ethical approval The authors confirm that all the research meets ethical guidelines.

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